

CIAN ROCHE

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EDUCATION

Massachusetts Institute of Technology 2021 – Present

Ph.D. Candidate, Physics

Thesis: “*The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector*”

Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger

Universität Tübingen 2019 – 2021

M.Sc., Mathematical Physics, 1.1 GPA[†]

Thesis: “*Exact parallel waves in general relativity*”

Advisor: Prof. Carla Cederbaum

University College Cork 2015 – 2019

B.Sc., Physics, 1st class honours

HONORS AND AWARDS

MIT Physics Henry W Kendall Teaching Fellowship 2026
DAAD Graduate Program Full Scholarship 2019
University College Cork College Scholar 2019
Max Planck Institute for Plasma Physics Research Scholarship 2018
Naughton Research Fellowship (one of 8 in Ireland) 2017
Naughton Foundation Scholarship (one of 29 in Ireland) 2015
All-Ireland Scholarship (one of 125 in Ireland; declined) 2015

TEACHING

Massachusetts Institute of Technology

Teaching Assistant, Experimental Physics II (8.14) Spring 2026
Teaching Assistant, Experimental Physics I (8.13) Fall 2024
Teaching Assistant, Experimental Physics I (8.13) Spring 2024
Teaching Assistant, Experimental Physics I (8.13) Fall 2023
Teaching Assistant, Experimental Physics I (8.13) Fall 2022
Instructor - Gaia DR3 Hackathon June 2022

University College Cork

Teaching Assistant, Introductory Physics II Spring 2018
Leader, Peer Assisted Learning Scheme 2016 – 2018

PUBLICATIONS

ADS library: <https://tinyurl.com/CianRocheADS>

43 first-author citations, 125 citations, *h*-index 6

First Author or Significant Contribution

7. **Roche, C.**, Dutra, I., McDonald, M., Meneghetti, M., Natarajan, P., Sharon, K., Vogelsberger, M., 2026; “*Host-Dependence and Line Of Sight Effects on Galaxy-Galaxy Strong Lensing in Clusters*”, (in prep)
6. Dutra, I., Natarajan, P., **Roche, C.**, 2026; “*Where Cluster Galaxies Reside and How Strongly They Lens*”, (in prep)
5. **Roche, C.**, Vogelsberger, M., McDonald, M., Dutra, I., Natarajan, P., Shen, X., Shreeram, X., Metcalf, R., Sharon, K., Birrer, S., Lee, W., Meneghetti, M. 2026; “*A Consistent Implementation of Cluster Strong Lensing in Cosmological Simulation Light Cones*”, accepted at OJA, [arXiv:2605.30433](https://arxiv.org/abs/2605.30433)

[†]on a scale of 1-5, where 1 is the highest grade.

4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O’Niell, S., **Roche, C.**, Zier, O., Gao, J. 2024; “*Novel Density Profile for Isothermal Cores of Dark Matter Halos*”, Phys. Rev. D 113, 063024, [arXiv:2411.11945](#)
3. **Roche, C.**, McDonald M., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; “*Brightest Cluster Galaxy Offsets in Cold Dark Matter*”, OJA, astro-ph.GA 7, [arXiv:2402.00928](#)
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; “*The Escape Velocity Profile of the Milky Way from Gaia DR3*”, ApJ, 972, 70, [arXiv:2402.00108](#)
1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; “*Exact parallel waves in general relativity*” Gen Relativ Gravit 55, 40, [doi:10.1007/s10714-023-03083-x](#)

Nth Author

3. Rose J. C., et al. (incl. **Roche, C.**) 2025; “*The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties*”, [arXiv:2512.00148](#)
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; “*How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds*”, ApJ, 997, 336, [arXiv:2409.02980](#)
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; “*Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations*”, ApJ, 982, 68, [arXiv:2405.00766](#)

Non-Refereed

1. **Roche, C.** & Garnavich, P. 2020; “*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*” Res. Notes AAS 4 207, [doi:10.3847/2515-5172/abcb05](#)

PRESENTATIONS

Invited (Colloquia/Seminars)

Talk	“ <i>Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations</i> ” MIT Kavli Institute for Astrophysics and Space Research	Mar 2026
Talk	“ <i>Understanding Dark Matter with Simulated Strong Lensing</i> ” Yale University	Sep 2025
Talk	“ <i>Dark Matter Self-Interaction via Galaxy Cluster Dynamics</i> ” Harvard University	Feb 2024
Talk	“ <i>Galaxies Lacking Dark Matter Produced in a Cosmological Simulation</i> ” MIT Kavli Institute for Astrophysics and Space Research	Apr 2022
Lecture	“ <i>Waves in General Relativity</i> ” Mathematical Relativity Class – University of Tübingen	July 2021
Talk	“ <i>Exact parallel waves in general relativity</i> ” Mathematical Physics Colloquium – University of Tübingen	June 2021
Talk	“ <i>How to Stop a Hurricane: Complexity in Physics</i> ” “Blow Your Mind Week” 2020 – University College Cork	Oct 2020
Talk	“ <i>Hermiticity of Young Operators</i> ” Young Tableaux Seminar Series – Universität Tübingen	June 2020

Contributed

Poster	“ <i>Galaxy-Scale Dark Matter Inference in Cluster Environments</i> ” Probe Combination for Dark Matter Physics in the Era of Large Surveys – Kavli Institute for Cosmological Physics, University of Chicago	Sep 2026
Poster	“ <i>The Line of Sight as a Major Systematic of Cluster Strong Lensing</i> ” Gravitational Lensing at the Turning Point – Sexten Center for Astrophysics	July 2026
Talk	“ <i>The Observation-Simulation Connection and the Line of Sight in Cluster Strong Lensing</i> ” Strong Gravitational Lensing Workshop – University of Montpellier	June 2026
Talk	“ <i>Consistent Strong Lensing Predictions from Cosmological Simulations</i> ” Scaling up Lensing Workshop – University of Liège	Oct 2025
Poster	“ <i>Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” Tracing cosmic Evolution with Galaxy Clusters V – Sexten Center for Astrophysics	July 2025
Talk	“ <i>Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” European Astronomical Society Meeting 2025 – University College Cork	June 2025

ADVOCACY AND OUTREACH

MIT GAGA Initiative Co-chair 2022 – 2023 | Organizing Committee 2021 – 2024
The Graduates Advising Graduate Admissions program offers information and mentorship to underrepresented groups in physics, and advises on improvements to the admissions process.

MIT PhysGAAP program 2021 – 2023
The Physics Graduate Application Assistance Program provides guidance to prospective applicants, and liaises with physics department leadership on issues of accessibility in the graduate application. I have acted as one of the primary organizers and have mentored 5 students through the program.

Astrogazers, Massachusetts Institute of Technology 2024–Present
Student academic representative, University College Cork 2017 – 2019

LEADERSHIP AND SERVICE

MIT Physics Graduate Admissions 2023 Cycle, 2024 Cycle
Served for two cycles as a graduate member of the physics graduate admissions committee.

MIT Summer Research Program Admissions Committee 2023 Cycle
The MIT Summer research program (MSRP) aims to provide high-quality research experiences to promising undergraduate students from underrepresented backgrounds.

Volunteer, AAS 241st Meeting Graduate School Fair 2023
Graduate Student Office Manager, MIT Kavli Institute 2022–2025
Organizer, MIT Kavli Institute Graduate Student Lunch Series 2022–2023
Member, MIT Physics Graduate Student Council 2021 – 2023
Organizing Committee, MIT Graduate Student Union 2021 – 2023
Astronomy Officer, UCC Physics and Astronomy Society 2017–2019

EDUCATIONAL RESOURCES

Caustica

An educational and intuition-building online strong lensing simulator, also capable of producing high-quality lensing media such as video and vector graphics.

χ by Eye

A learning tool in a geoguessr-like format for training students to recognize believable or unrealistic statistics in a plot by eye.

AstroWiki

Created and now co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam.